COURSE NUMBER: MKTG 4163 / 6253 – Section 80 CRN: 45312 / 45313
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Spring 2026 – Thursdays – 7:10 PM-9:40 PM – Duques Hall 353

COURSE TITLE: Applied Marketing Analytics

Course Objectives

The role of marketing and general business analytics has become increasingly important in marketing decision-making and has in fact grown substantially in the past 10 years. *Forbes* magazine reported a 67-percent growth in marketing analytics hires during the previous year and an amazing 136-percent growth during the previous three years. Levering on this increasing growth in marketing analytics, this course objectives are to teach students how to merge marketing and analytics using various tools to include statistical software to analyze qualitative & quantitative primary & secondary data to support marketing decisions. Student teams will report their analytic results with a final briefing at the end of the semester. There will be ongoing hands-on case examples and team exercises to apply the concepts in a marketing context.

Course Overview and Learning Objectives

Well run organizations use marketing analytics to improve their understanding of the internal and external factors that affect their business. This course will focus on an applied approach to learning about transforming data into actionable information used in making informed marketing decisions. Student teams analyze primary qualitative and quantitative data in support of a marketing objective. Students will learn a practical application to data analytics through data drills in a data lab setting. Using EXCEL based statistical software students will learn how to identify data nuances and to weave qualitative and quantitative data into a compelling story.

Students Will Learn How To

- Using analytics to tell the marketing story based on data;
- Manage and analyze big data for marketing decision making;
- Deal with data quality, sufficiency, and relevance issues;
- Use statistical software to analyze data using descriptive and inferential data analysis techniques;

Materials

- Optional reference: Marketing Analytics by Wayne Winston (Wiley Publishing) – ISBN-13: 978-1118373439
- Various datasets and handouts will be used as case examples for team exercises
- Articles on contemporary issues related to marketing analytics

Grading

- Exam I and II 20%, Attendance/Active Class and Project Participation 20%, Project 30%, Final 10%

Class Schedule

1/15: Introduction to Marketing Analytics

- Course overview
- Data storytelling
- Difference between performance measures and marketing metrics
- Introduction to marketing analytics and its use in marketing decision-making

1/22: Using Excel to Summarize Marketing Data

Excel Tools:

- *Simpson's Paradox*
- *Descriptive statistics*
- *Rank and Percentile*
- *Multivariate pivot tables*
- *Data slicers and dashboards*

1/29: Using Excel Functions to Tabulate Data and Types of Data Models

Excel Tools:

- *Average/count if and average/count ifs functions*
- *V-lookup tables*
- *X-lookup and index match tables*

2/05: Price Bundling and Optimal Price & Quantity Calculations

Excel Tools:

- *Price bundling calculations*
- *Using Solver to optimize outcomes*
- *Strategy Canvas*

2/12: Testing of Differences and Forecasting

- *Z-test and Chi Square*
- *Multivariate regression*
- *Monte Carlo simulation*

2/19: Exam I *(1st half of class)*

AI, Forecasting and Queuing Theory

Excel Tools:

- Essays and data exercises
- Using artificial intelligence in analytics
- *Conjoint analysis & exponential smoothing*

2/26: Trend Analysis and Path Dependency

Excel Tools:

- *Trend function*
- *Holt-Winters forecasting*
- *Moving averages*
- *Probability functions*

3/05: Network Effects and Understanding Customer Wants

Excel Tools:

- *Network effects*
- *Implications wheel*
- *Combination vs. permutations*
- *Using weight effects to predict variables importance*
- *Connecting data points to produce patterns*

3/12 – No Class – Spring Break

3/19: Using randomness to predict marketing effectiveness

Excel Tools:

- *Markov Chains*
- *The wisdom of crowds*
- *Using random number generator functions to validate samples*

3/26: Market Segmentation

Excel Tools:

- *Decision trees*
- *Implications wheel*
- *Stem and box plots*

4/02: Exam II *(1st half of class)*

- Essays and data exercises
- Project work for the second half of the class

4/09: Measuring Advertising Effectiveness and Variance Analysis

Excel Tools:

- *Histograms*
- *Correlation and coefficients of determination*
- *ANOVA – single factor, two factor with and without replication*

4/16: Catch up and Presentation Preparation

- Review key concepts
- Work on presentation briefs

4/23: Team Presentations – 12 minutes plus 3 minutes for questions – total 15

TBD – Final Exam Date/Location

Team Project

Students will select a dataset from Kaggle, or another source, to analyze using a variety or tools we discuss in class. The focus should be on tell the story of what their data set is saying. Students will submit a 25-page paper and presentation to the class of their research findings. The teams should include other sources of primary and secondary to enhance their analysis. A general outline of the project should include:

- Project and team introduction (*for the team presentation*)
- Dataset context and background
- Research objectives
- Project methods: Incorporation of other primary and secondary sources
- Data analysis using the tool discussed in class
- Concluding and recommendations on what to next with the data
- Data and project limitations
- "Thank You" Q&A (*for the team presentation*)

Some things to consider when building your report:

Data analysis is a critical process in extracting valuable insights from raw data, helping organizations and individuals make informed decisions. Whether it's for academic research, business strategy, or product development, effective data analysis is essential for understanding trends, patterns, and relationships within the data. To ensure the analysis is thorough, accurate, and meaningful, certain key steps and considerations must be followed.

1. Defining the Problem or Objective – and Provide Data Context
Before diving into the data, it is crucial to clearly define the research question or problem to be solved. A focused objective helps in determining the type of data required and the appropriate methods for analysis. This step ensures that the analysis is aligned with the goals of the project, whether it is to identify patterns, test hypotheses, or predict future outcomes.

2. Data Collection and Preparation – Distinguish between Primary and Secondary
Once the problem is defined, the next step is gathering relevant data. This may involve collecting new data through surveys, experiments, or observations, or sourcing existing data from databases or other secondary sources. The quality and relevance of the data are paramount, as unreliable or irrelevant data can lead to inaccurate conclusions.
After collection, the data often requires cleaning and preparation. This includes handling missing values, correcting errors, and transforming the data into a usable format. Data cleaning ensures that the analysis is based on accurate and consistent information. Additionally, this stage may involve standardizing units, encoding categorical variables, or filtering out outliers that could distort results.

3. Exploring the Data
Exploratory Data Analysis (EDA) is an essential phase where analysts use statistical and graphical techniques to explore the data before applying complex models. Visualizations like histograms, scatter plots, and box plots, as well as summary statistics such as mean, median, and standard deviation, help in understanding the data's distribution, relationships, and anomalies. EDA helps in detecting patterns or trends that might guide further analysis and modeling.

4. Selecting the Appropriate Analysis Methods
Using a variety of the tools we discuss in class – the choice of analysis method depends on the type of data and the objective of the analysis. Match the analysis technique with the research question, as the wrong approach can lead to misleading results. Common techniques include:

- **Descriptive analysis**, which summarizes the main features of the dataset – the who, what, where, when how – charts/graphs/sums/averages/counts/percentages.

- **Inferential statistics**, which allows making generalizations about a population based on sample data. Understanding the "why" of the data. Correlation, ANOVA, and Chi Square analyses are useful here.

- **Predictive modeling**, such as regression analysis, or machine learning techniques, which forecast future outcomes based on historical data. Tools you might use include Holt-Winters, Markov Chains, Queueing Models, and Monte Carlo Simulations.

- **Prescriptive analysis**, which provides recommendations based on the findings. A combination of the earlier tools can help you to pull together recommendations.

5. Interpretation and Conclusion
After performing the analysis, the next critical step is interpreting the results in the context of the original research question. This involves making sense of the statistical findings and drawing meaningful conclusions. It is important to evaluate the results carefully, considering possible limitations and biases in the data. Additionally, analysts should assess the validity and reliability of their findings before presenting them.

6. Communication of Results
Finally, communicating the results effectively is crucial. This can involve preparing reports, dashboards, or presentations tailored to the audience, whether they are business executives, policymakers, or academics. Clear visualization of key findings and a concise summary of the implications of the analysis can make the results actionable and impactful.

7. Conclusion
Conducting data analysis is a systematic process that requires a clear objective, careful data collection and preparation, exploratory analysis, thoughtful selection of methods, and clear communication of findings. By following these essential steps, analysts can transform raw data into valuable insights that drive informed decisions and improve outcomes across various fields.